DATASHEET

# ChArUco Calibration Target, Aluminum Composite

ChArUco combines a checkerboard with ArUco markers in the white squares. Markers identify each corner, so the board can be partly outside the image or partly occluded and still contribute corners — which is exactly where lens distortion is strongest. Native in OpenCV (`cv2.aruco.CharucoBoard`), supported by MATLAB and most multi-camera toolchains.

## 1 Ordering

| Part number | Size (W × H) | Squares (rows × cols) | Square / marker | Dictionary | Corners | Markers |
|---|---|---|---|---|---|---|
| CT-CHA-P-200X150-{DICT} | 200 × 150 mm | 5 × 7 | 25 / 18 mm | see below | 24 | 17 |
| CT-CHA-P-300X200-{DICT} | 300 × 200 mm | 7 × 10 | 25 / 18 mm | | 54 | 35 |
| CT-CHA-P-400X300-{DICT} | 400 × 300 mm | 9 × 12 | 30 / 22 mm | | 88 | 54 |
| CT-CHA-P-600X400-{DICT} | 600 × 400 mm | 9 × 14 | 40 / 30 mm | | 104 | 63 |
| CT-CHA-P-800X600-{DICT} | 800 × 600 mm | 11 × 15 | 50 / 37 mm | | 140 | 82 |

`{DICT}`: D6 = DICT_6X6_250 (default, most robust) · D5 = DICT_5X5_100 · D4 = DICT_4X4_50 (decodes at lower resolution, fewer unique IDs) · T36 = AprilTag 36h11. Start ID 0, row-major over white squares, OpenCV ≥ 4.6 placement convention (`legacy_pattern: false`). Other grids, dictionaries or start IDs: custom order (CUS).

Pattern area is centered with a white quiet zone ≥ 1 square on all sides; the edge strip with part number, serial and 100 mm scale bar sits outside the quiet zone.

## 2 Key specifications

ACCURACY

±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C)

SUBSTRATE

Aluminum / LDPE / aluminum composite, 6 mm

FLATNESS

≤ 0.5 mm per meter, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

## 1 Specifications · geometry

| FEATURE | SPECIFICATION |
|---|---|
| Feature position tolerance | ±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C) |
| Square pitch, mean | ±0.05 mm |
| Orthogonality | ±0.02° |
| Marker-to-square ratio | 0.72–0.75 (see ordering table for per-size marker dimensions) |
| Edge radius | R6 mm |

## 2 Specifications · physical

| FEATURE | SPECIFICATION |
|---|---|
| Substrate | Aluminum / LDPE / aluminum composite, 6 mm |
| Weight | 6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg) |
| Pattern | Direct UV print, matte, black on white; optical density ≥ 1.6 |
| Flatness | ≤ 0.5 mm per meter, board supported on three points |
| Rigidity (E·J) | ≈ 3800 kN·cm²/m |
| Mounting | 4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size |

## 3 Specifications · environment

| FEATURE | SPECIFICATION |
|---|---|
| Operating temperature | −50 to +80 °C |
| Thermal expansion | ≈ 24 µm/(m·°C) |
| Storage | −20 to +60 °C, dry |
| Outdoor lifetime | up to 3 years (UV-stable print) |
| Cleaning | water or ≤ 70 % isopropyl alcohol, soft cloth |

## 4 Deliverables

| INCLUDED | | SERVICE | |
|---|---|---|---|
| Measurement | Every board is measured before it ships (dimensional, flatness, visual) | Lead time | Made to order · ships in 2–3 weeks |
| Files | `.opencv.json`, `.ros2.yaml`, `.halcon.txt` (HALCON point tuple), `.dxf` | Options | Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE |
| Documents | this datasheet, CT-UM-GEN user manual, CT-UM-CHA-P quick start | | |

## 5 Compliance & customs

| TOPIC | STATEMENT |
|---|---|
| RoHS | RoHS-compliant materials. |
| HS code | `9031 80` Not a toy. not for food contact. |

## 6 Revision history

| REV | DATE | CHANGE |
|---|---|---|
| A | 2026-10 | First issue, planning values |

## 7 Disclaimer

Values are nominal unless stated as measured.